



Installing Deepstream on Nvidia Jetson AGX Orin AIR-030

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Prerequisites

Check the following specifications before starting your installation process.

We are using Deepstream 7.0, needed specifications are:

- Compatible CUDA-version
- Compatible NVIDIA GPU

You can check these specifications using either **jtop** (if installed):

```
jtop 4.3.1 - (c) 2024, Raffaello Bonghi [raffaello@rnext.it]
Website: https://rnext.it/jetson_stats

Platform
Machine: aarch64
System: Linux
Distribution: Ubuntu 22.04 Jammy Jellyfish
Release: 5.15.136-tegra
Python: 3.10.12

Serial Number: [s]XX CLICK TO READ XXX
Hardware
Model: NVIDIA Jetson AGX Orin Developer Kit
699-level Part Number: 699-13701-0005-500 R.0
P-Number: p3701-0005
Module: NVIDIA Jetson AGX Orin (64GB ram)
SoC: tegra234
CUDA Arch BIN: 8.7
L4T: 36.3.0
Jetpack: 6.0

Libraries
CUDA: 12.2.140
cuDNN: 8.9.4.25
TensorRT: 8.6.2.3
VPI: 3.1.5
Vulkan: 1.3.204
OpenCV: 4.12.0-dev with CUDA: YES

Hostname: localhost.localdomain
Interfaces
eth01: 172.21.96.63

1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7INFO Quit (c) 2024, RB
```

Or **nvidia-smi**:

```
ubuntu@localhost:~$ nvidia-smi
Mon Apr 7 11:56:27 2025

+-----+
| NVIDIA-SMI 540.3.0 | Driver Version: N/A | CUDA Version: 12.2 |
+-----+-----+
| GPU  Name          | Persistence-M | Bus-Id  Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf    | Pwr:Usage/Cap |      Memory-Usage | GPU-Util  Compute M. |
|                      |                |                 |      GPU-MIG  M.    |
+-----+-----+
|  0  Orin (nvgpu)    |      N/A      | N/A     N/A    |           N/A      |
| N/A   N/A   N/A     | N/A /  N/A    | Not Supported |           N/A      |
|                      |                |                 |           N/A      |
+-----+-----+

+-----+
| Processes:         |
| GPU  GI    CI       | PID   Type   Process name      | GPU Memory |
| ID   ID   ID       |       |       |                   | Usage      |
+-----+-----+
| No running processes found |
+-----+-----+

```

Installation

Step 1: Download Deepstream 7.0

Find the right package:

Go to the Deepstream SDK page and find Deepstream 7.0.

<https://catalog.ngc.nvidia.com/orgs/nvidia/resources/deepstream/version>

Download .deb file:

Our Nvidia Jetson AGX Orin AIR-030's architecture is ARM

Choose deepstream-7.0 7.0.0-1 arm64.deb

Overview	Version History	File Browser	Release Notes	Related Collections
Version 7.0 Search files...				
File	Size	Modified	Actions	
 deepstream-7.0_7.0.0-1_amd64.deb	767.79 MB	a year ago	...	
 deepstream-7.0_7.0.0-1_arm64.deb	630.70 MB	a year ago	...	
 deepstream_libraries-1.0-cp310-cp310-linux_x86_64.whl	153 kB	a year ago	...	
 deepstream_sdk_v7.0.0_jetson.tbz2	655.90 MB	a year ago	...	
 deepstream_sdk_v7.0.0_x86_64.tbz2	830.46 MB	a year ago	...	

Step 2: Install Deepstream 7.0

Locate and install:

- Navigate to the directory where you downloaded the .deb package:

```
ubuntu@localhost:~$ cd Downloads/
ubuntu@localhost:~/Downloads$ ls
AIR-030.pdf                                pexels-dreamypixel-547115.jpg
cuda-keyring_1.1-1_all.deb                 pexels-life-of-pix-7919.jpg
deepstream-7.0_7.0.0-1_arm64.deb          pexels-souvenirpixels-1519088.jpg
Edge_AI_SDK-installer-2.0.1.tar.gz        pexels-thanhhoa-tran-640546-1447092.jpg
Edge_AI_SDK-installer-3.0.0              torch-2.3.0-cp310-cp310-linux_aarch64.whl
Edge_AI_SDK-installer-3.0.0.tar.gz        torchvision-0.18.0a0+6043bc2-cp310-cp310-linux_aarch64.whl
martin-martiz-1hg6Np00kIk-unsplash.jpg   vmlinuz
MobileNet-v2.onnx                        webots_2025a_amd64.deb
ngcccli_arm64.zip                       yolov3-tiny.weights
onnxruntime_gpu-1.19.0-cp310-cp310-linux_aarch64.whl
```

- Install the .deb file using dpkg:

```
ubuntu@localhost:~/Downloads$ sudo dpkg -i deepstream-7.0_7.0.0-1_arm64.deb
```

- Install any missing dependencies:

```
ubuntu@localhost:~/Downloads$ sudo apt-get install -f
```

- Verify installation:

```
ubuntu@localhost:~/Downloads$ deepstream-app --version
deepstream-app version 7.0.0
DeepStreamSDK 7.0.0
```

Step 3: Test Deepstream

Navigate and run:

Navigate to the Deepstream sample directory:

```
ubuntu@localhost:~$ cd /opt/nvidia/deepstream/deepstream-7.0/samples/
ubuntu@localhost:/opt/nvidia/deepstream/deepstream-7.0/samples$
```

Run sample application

```
ubuntu@localhost:/opt/nvidia/deepstream/deepstream-7.0/samples$ deepstream-app -c configs/deepstream-app/source1_usb_dec_infer_resnet_int8.txt
```